

③ Yes The time required for the Monte Carlo simulation to sample all states was estimated using multiple independent runs for each sequence. The average number of steps and associated uncertainties show a clear dependence on sequence.

The all P P P P P P P P sample all states the fastest, requiring approximately $(2.85 \pm 0.11) \times 10^4$ steps, while all 'H' H H H H H H H is the slowest, requiring about $(2.55 \pm 0.12) \times 10^5$ steps. The mixed HP sequences fall in between with sampling times into range of approximately $(6.7 - 7.5) \times 10^4$ steps.

This shows that sampling is strongly sequence dependent. Also this shows hydrophobic rich sequences take longer to fully explore all the configurations than the polar sequence. This can be due to low energy structures that restrict movement and slow down exploration.